WAREHOUSE MANAGEMENT SYSTEM

SDLC Documentation

HLD

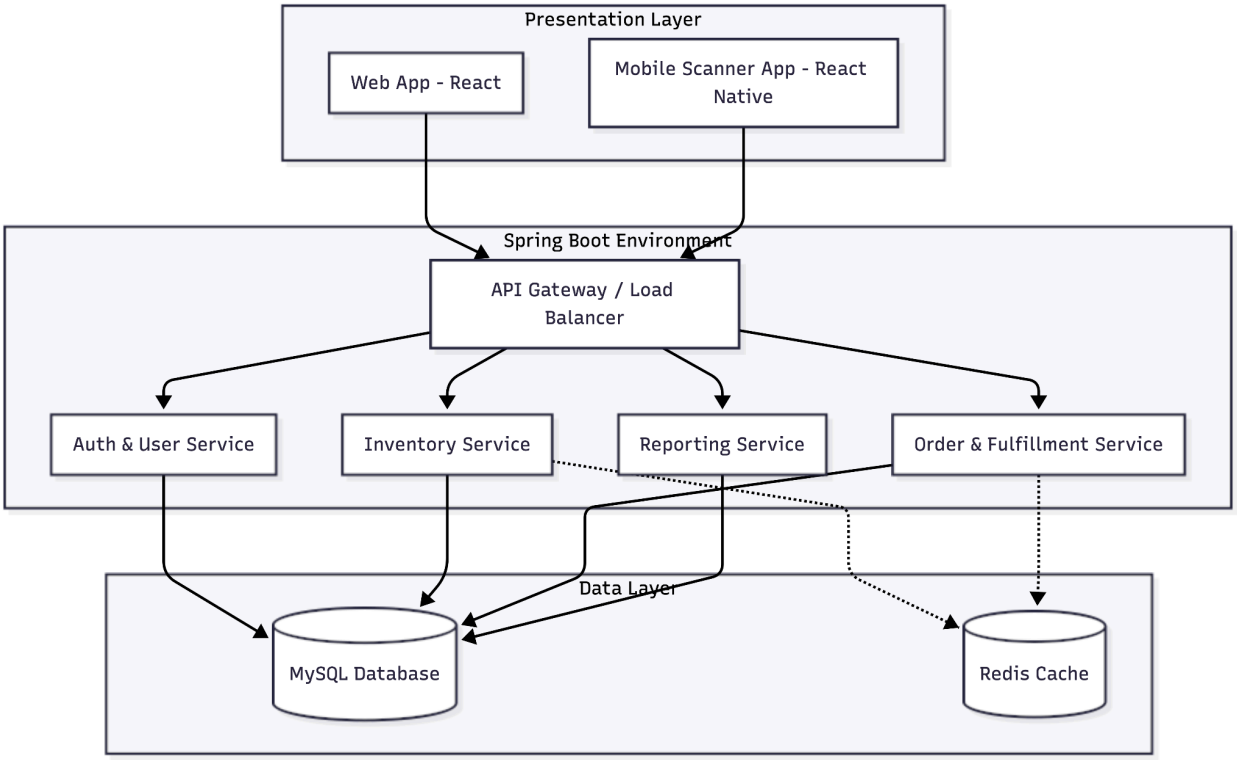
Field	Details
Project Name	Warehouse Management System
Technology	Java Spring Boot
Prepared By	Team Mavericks

HLD – High Level Design

This section describes the overall architecture, technology stack, system components, and data model of the Warehouse Management System(WMS) from a high-level perspective.

3.1 System Architecture

The WMS follows a classic three-tier RESTful architecture:



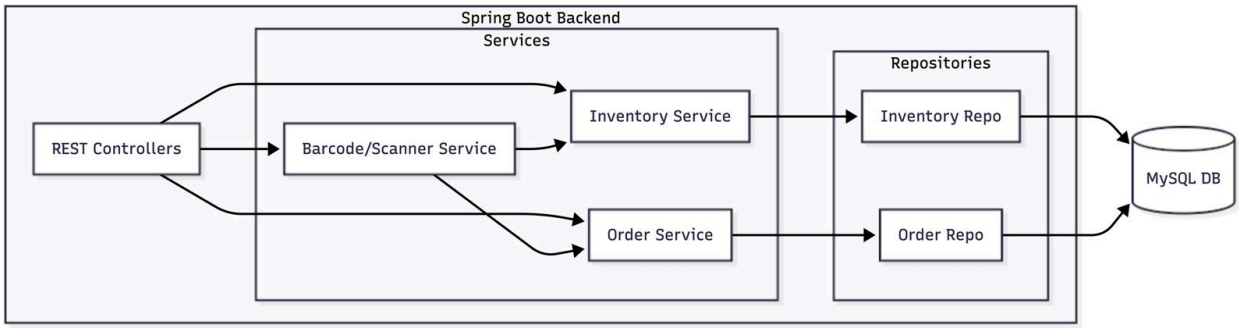
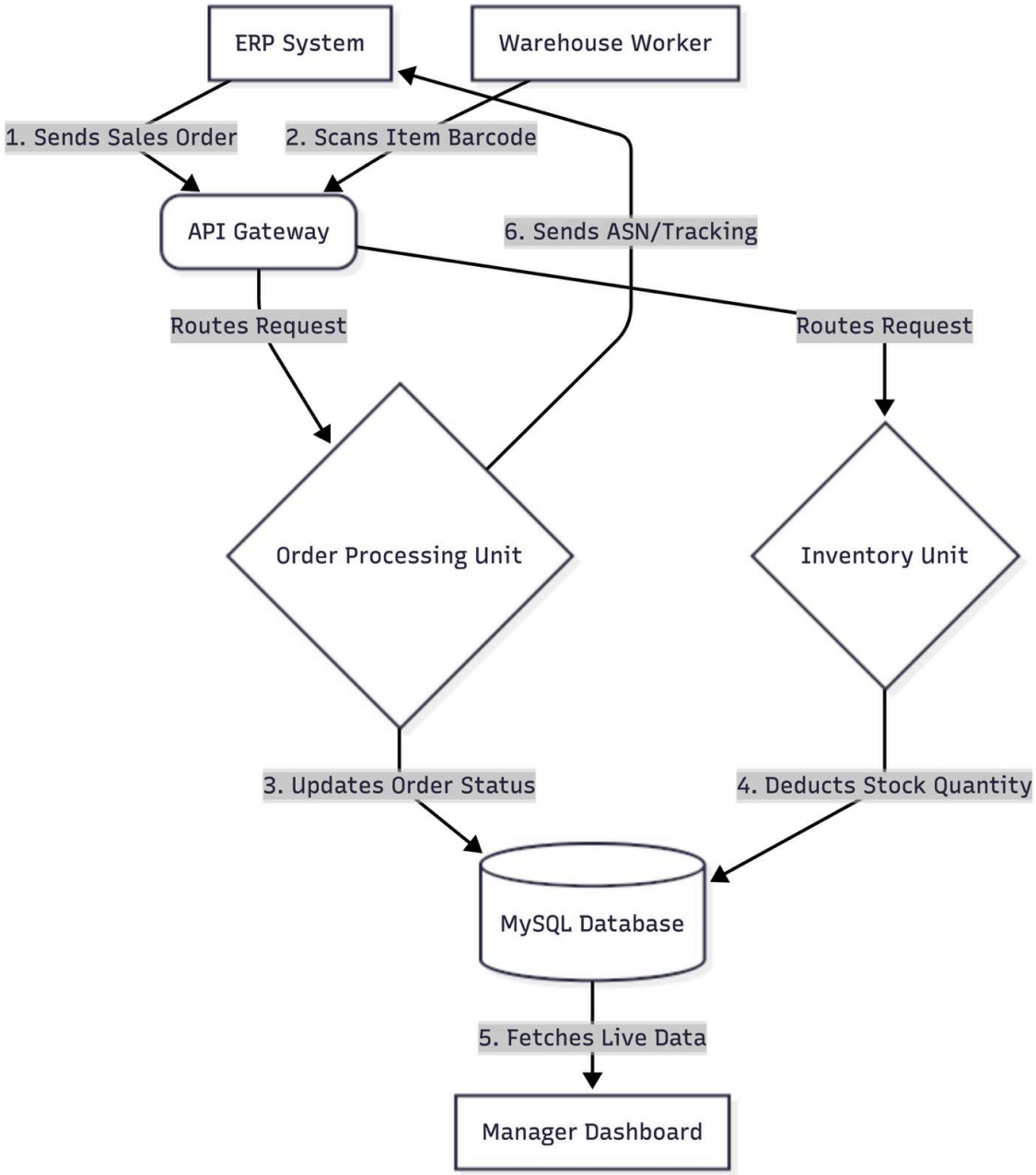
3.2 Technology Stack

Layer	Technology	Justification
Backend Framework	Java Spring Boot	Production-grade REST API, embedded Tomcat
Language	Java 17	Stability, strong typing, enterprise ecosystem
Database	MySQL	ACID transactions, relational integrity, JSON support
ORM	Spring Data JPA / Hibernate	Simplified DB operations, entity mapping
Security	Spring Security + JWT	Stateless auth, role-based access control
Build Tool	Maven / Gradle	Dependency management and build automation
API Testing	Postman / Swagger UI	API documentation and manual testing
Version Control	Git / GitHub	Source code management and collaboration
Deployment	Docker + Cloud (AWS)	Containerized, scalable deployment

3.3 System Components

Component	Responsibility
REST Controllers	Handle HTTP requests, validate inputs, return structured JSON responses.
Service Layer	Business logic: stock calculation, validation rules, transaction management.
Repository Layer	Database access via Spring Data JPA; custom queries for reports and filtering.
Entity Models	JPA entity classes mapping to PostgreSQL tables (Item, Shipper, ShippingIn, ShippingOut).
DTO Layer	Data Transfer Objects for request/response decoupling from entity models.
Exception Handler	Global @ControllerAdvice to return consistent error responses with HTTP codes.
Security Module	JWT token validation, filter chain, and role-based endpoint protection.

3.4 Component Diagram:



3.5 Dataflow Diagram: